

Partial Identification of Average Treatment Effect Under Covariate Contamination

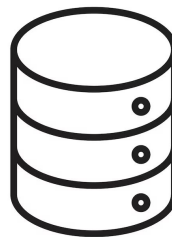
How much can we learn about treatment effects from contaminated covariate data? We use reliable auxiliary data to bound plausible effects and quantify uncertainty.

clean dataset



(X, A)

contaminated dataset



$(\tilde{X}, A, Y)$

Estimand: Average Treatment Effect

clean dataset

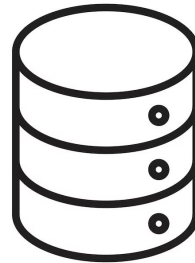


$$(X, A) \sim P_0$$

$$P^0(X = x \mid A = a) = P^1(X = x \mid A = a)$$

Distributional compatibility

contaminated dataset



$$(\tilde{X}, A, Y) \sim P_1$$

$$P^1(X \neq \tilde{X} \mid A = a, Y = y) \leq \rho_{a,y}$$

Sensitivity parameter

Require:

- X discrete, A and Y binary
- $P^1(X \neq \tilde{X} \mid A = a, Y = y) \leq \rho_{a,y}$ ●
- $P^0(X = x \mid A = a) = P^1(X = x \mid A = a)$ ●

Do not require:

- linkage of the records from the two datasets
- the two datasets contain same entries

clean dataset

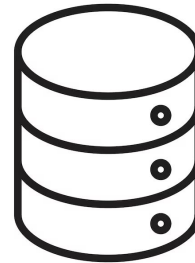


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Do not require:

- linkage of the two dataset
- the two dataset to contain same entries

$$P^1(X \neq \tilde{X} \mid A = a, Y = y) \leq \rho_{a,y}$$

- intuitive for interpretation
- not amicable for optimization

$$D_{\text{TV}}\left(P_{X|A=a,Y=y}^1, P_{\tilde{X}|A=a,Y=y}^1\right) \leq \rho_{a,y}$$

- not intuitive for interpretation
- amicable for optimization

Fact:
$$D_{\text{TV}}\{P^1(X \mid A = a, Y = y), P^1(\tilde{X} \mid A = a, Y = y)\} \leq P^1(X \neq \tilde{X} \mid A = a, Y = y).$$

Proposition: the partial identification bound resulting from those sensitivity parameters definition result in the same population level partial identification interval.

Theorem 4 (Sharp identification of the ATE) Suppose that $P^1(Y = y \mid A = a) > 0$ for every $(a, y) \in \{0, 1\}^2$. Under Assumptions 1–3, the sharp lower bound on the ATE is

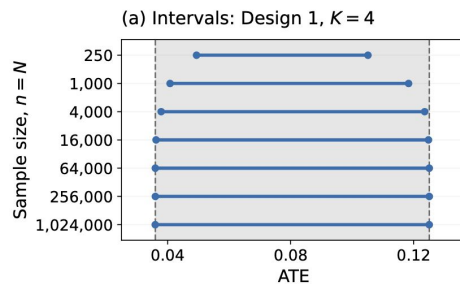
$$\begin{aligned} \theta_L &= \min_{m_0, m_1} \sum_{x \in \mathcal{X}} p(x) \{m_1(x) - m_0(x)\} \\ &\quad \text{subject to} \\ 0 &\leq m_a(x) \leq 1, \quad \forall a \in \{0, 1\}, x \in \mathcal{X}, \\ \sum_{x \in \mathcal{X}} P^1(X = x \mid A = a) m_a(x) &= P^1(Y = 1 \mid A = a), \quad \forall a \in \{0, 1\}, \\ D_{\text{TV}} \left(\frac{P^1(X = \cdot \mid A = a) m_a(\cdot)}{P^1(Y = 1 \mid A = a)}, P^1(\tilde{X} = \cdot \mid Y = 1, A = a) \right) &\leq \rho_{a,1}, \quad \forall a \in \{0, 1\}, \\ D_{\text{TV}} \left(\frac{P^1(X = \cdot \mid A = a) \{1 - m_a(\cdot)\}}{P^1(Y = 0 \mid A = a)}, P^1(\tilde{X} = \cdot \mid Y = 0, A = a) \right) &\leq \rho_{a,0}, \quad \forall a \in \{0, 1\}. \end{aligned}$$

The auxiliary data provides two major benefits:

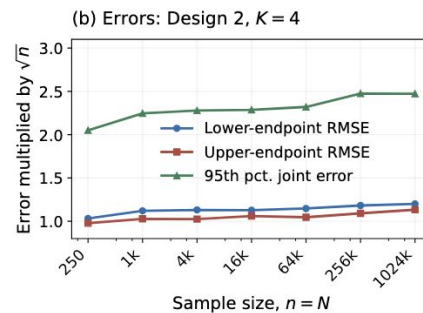
- 1) can significantly reduce width the population partial-identification bounds
- 2) makes partial identification a *linear* program
- 3) allows root-n consistency identification interval

K	With auxiliary data	Width	Without auxiliary data	Reduction
2	[0.0475, 0.1125]	0.0650	$[-0.0016, 0.1400]$	54.1%
4	[0.0360, 0.1251]	0.0891	$[-0.0713, 0.1603]$	$\geq 61.5\%$
16	[0.0329, 0.1351]	0.1022	$[-0.0980, 0.2016]$	$\geq 65.9\%$

width reduction using auxiliary data



consistency



root-n consistency